

Cool Project

We cooked with Toeplit matrices today following Martha Stewart

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Contents

1	Introduction	2
2	Definitions	2
2.1	Toeplitz Matrices	2
2.1.1	Quasi Toeplitz Matrices	2
2.2	Displacement Matrices	2
2.2.1	Mathematical Approach	2
2.2.2	Computational Approach	3
2.3	Generator Matrices	3

1 Introduction

Hello hello welcome!

2 Definitions

2.1 Toeplitz Matrices

A Toeplitz matrix is in which each element in first row and first column descends diagonally. For a matrix $A \in \mathbb{K}^{n \times m}$, it satisfies the condition $A_{i,j} = A_{i+1,j+1}$. Formally, it is defined by a sequence of elements $\{a_k\}$ such that:

$$A = \begin{pmatrix} a_0 & a_{-1} & a_{-2} & \cdots & a_{-(m-1)} \\ a_1 & a_0 & a_{-1} & \cdots & a_{-(m-2)} \\ a_2 & a_1 & a_0 & \cdots & a_{-(m-3)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n-1} & a_{n-2} & a_{n-3} & \cdots & a_{n-m} \end{pmatrix} \quad (1)$$

$$A_{i,j} = a_{i-j}, \quad 0 \leq i < n, \quad 0 \leq j < m$$

2.1.1 Quasi Toeplitz Matrices

A Quasi-Toeplitz matrix is not strictly Toeplitz but shares similar structural properties. Can benefit the structural advantages of Toeplitz partially in these matrices.

2.2 Displacement Matrices

For a matrix $A \in \mathbb{K}^{n \times m}$, we define the displacement matrix ∇A using:

$$\nabla A = A - ZAZ^T \quad (2)$$

where Z represents the lower shift matrix and Z^T represents the upper shift matrix. Essentially, Z is a matrix with ones on the first sub-diagonal and zeros everywhere else:

$$Z = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix} \quad Z^T = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \quad (3)$$

Multiplying a matrix A by Z from the left (ZA) shifts the rows of A down by one (zeros at the top). Multiplying by Z^T from the right (AZ^T) shifts the columns to the right, (zeros at the first column).

2.2.1 Mathematical Approach

Result, operation ZAZ^T shifts the entire matrix A one step to the bottom-right.

For a generic Toeplitz matrix of size $n \times m$, this subtraction results in zeros everywhere except for the first row and the first column.

$$\nabla A = \begin{pmatrix} y_0 & x_1 & x_2 & \cdots & x_{m-1} \\ y_1 & 0 & 0 & \cdots & 0 \\ y_2 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_{n-1} & 0 & 0 & \cdots & 0 \end{pmatrix} \in \mathbb{K}^{n \times m} \quad (4)$$

This allows us to represent the $n \times m$ matrix using only $O(n + m)$ elements rather than storing the full $O(nm)$ dense matrix.

2.2.2 Computational Approach

While this definition involves matrix multiplication, in this case $O(n^2 * m)$ or similar depending on the multiplication method used, calculating ∇A explicitly via matrix multiplication is inefficient.

The displacement calculation can be reduced to an element based loop with complexity $O(nm)$:

$$(\nabla A)_{i,j} = \begin{cases} A_{i,j} & \text{if } i = 0 \text{ or } j = 0 \\ A_{i,j} - A_{i-1,j-1} & \text{otherwise} \end{cases} \quad (5)$$

This approach allows us to iterate from the bottom row to the top updating values in place.

2.3 Generator Matrices

Displacement matrices, ∇A , typically have a low rank α (displacement rank). For Toeplitz matrices, α is generally small ($\alpha \leq 2$). This allows us to store efficiently the displacement matrix into two smaller matrices G and H .

While the product GH^T results in the displacement ∇A , the original matrix A can be reconstructed using the columns of the generators mentioned at Proposition 10.5 in [1].

Let g_k and h_k denote the k -th columns of G and H respectively.

The matrix A can be written as a sum of elements of lower and upper triangular matrices:

$$A = \sum_{k=1}^{\alpha} L(g_k)U(h_k) \quad (6)$$

- $L(g_k)$ is a lower triangular Toeplitz matrix with the first column g_k .
- $U(h_k)$ is an upper triangular Toeplitz matrix with the first row h_k^T .

References

- [1] Alin Bostan, Frédéric Chyzak, Marc Giusti, Romain Lebreton, Grégoire Lecerf, Bruno Salvy, and Éric Schost. Algorithmes efficaces en calcul formel, August 2017. 686 pages. Édition 1.0.